Equations of Lines in 3D: Vector Product Equations

Questions 1–10

Question 1. Find the vector product equation of the line:

a) that passes through $(2, 8, 1)$ and is parallel to $5\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$

b) that passes through $(-10, 3, -5)$ and is parallel to $7\mathbf{i} - \mathbf{k}$

Question 2. Show that the point $(2, 0, 3)$ lies on the line with equation

$$\left(\mathbf{r} - \begin{pmatrix} 6 \\ 3 \\ -2 \end{pmatrix}\right) \times \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix} = \mathbf{0}.$$

Question 3. Find a vector product equation of the line passing through:

a) $(-3, 0, 9)$ and $(5, 4, -2)$

b) $(-2, -1, -6)$ and $(-3, 9, 25)$

Question 4. Determine whether the following pair of lines are parallel or not:

$$l_1 : \left(\mathbf{r} - \begin{pmatrix} 3 \\ 9 \\ -12 \end{pmatrix}\right) \times \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix} = \mathbf{0}, \quad l_2 : \left(\mathbf{r} - \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix}\right) \times \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = \mathbf{0}.$$

Question 5. Determine whether the following pair of lines are parallel, skew or if they intersect. If they intersect, find the point of intersection:

a) $l_1 : \left(\mathbf{r} - \begin{pmatrix} 3 \\ -8 \\ 4 \end{pmatrix}\right) \times \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} = \mathbf{0}, \quad l_2 : \left(\mathbf{r} - \begin{pmatrix} 6 \\ 5 \\ -3 \end{pmatrix}\right) \times \begin{pmatrix} -4 \\ 3 \\ 1 \end{pmatrix} = \mathbf{0}$

b) $l_1 : \left(\mathbf{r} - \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix}\right) \times \begin{pmatrix} -1 \\ -5 \\ 1 \end{pmatrix} = \mathbf{0}, \quad l_2 : \left(\mathbf{r} - \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix}\right) \times \begin{pmatrix} 6 \\ 5 \\ -6 \end{pmatrix} = \mathbf{0}$

Question 6. Find the acute angle between:

$$l_1 : \left(\mathbf{r} - \begin{pmatrix} 3 \\ 0 \\ -8 \end{pmatrix}\right) \times \begin{pmatrix} 2 \\ -9 \\ 0 \end{pmatrix} = \mathbf{0}, \quad l_2 : \left(\mathbf{r} - \begin{pmatrix} 4 \\ -6 \\ -7 \end{pmatrix}\right) \times \begin{pmatrix} 7 \\ 6 \\ 1 \end{pmatrix} = \mathbf{0}.$$

Question 7. It is given that:

$$l_1 : \left(\mathbf{r} - \begin{pmatrix} 5 \\ -8 \\ 6 \end{pmatrix}\right) \times \begin{pmatrix} 3 \\ -7 \\ 1 \end{pmatrix} = \mathbf{0}, \quad l_2 : \left(\mathbf{r} - \begin{pmatrix} -9 \\ 10 \\ 3 \end{pmatrix}\right) \times \begin{pmatrix} -13 \\ 1 \\ -1 \end{pmatrix} = \mathbf{0}.$$

a) Show that the lines l_1 and l_2 intersect. Hence find the coordinates of the point of intersection.
b) Find the acute angle between the lines l_1 and l_2.
c) The point A has position vector $\mathbf{i} + 15\mathbf{k}$. B lies on l_1 such that AB is perpendicular to l_1. Find the coordinates of B.

Question 8. Write the following vector product equations of lines in the form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{b}$:

a) $\left(\mathbf{r} - \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}\right) \times \begin{pmatrix} 2 \\ 9 \\ 1 \end{pmatrix} = \mathbf{0}$

b) $\left(\mathbf{r} - \begin{pmatrix} 4 \\ 3 \\ -2 \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ 2 \\ -3 \end{pmatrix} = \mathbf{0}$

Question 9. Write the following vector product equations of lines in cartesian form:

a) $\left(\mathbf{r} - \begin{pmatrix} 3 \\ 2 \\ -9 \end{pmatrix}\right) \times \begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix} = \mathbf{0}$

b) $\left(\mathbf{r} - \begin{pmatrix} 8 \\ -1 \\ 0 \end{pmatrix}\right) \times \begin{pmatrix} -2 \\ -3 \\ 1 \end{pmatrix} = \mathbf{0}$

c) $\left(\mathbf{r} - \begin{pmatrix} 0 \\ 2 \\ 9 \end{pmatrix}\right) \times \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} = \mathbf{0}$

Question 10. Write the following vector product equations in both vector form and cartesian form:

a) $\left(\mathbf{r} - \begin{pmatrix} 3 \\ -2 \\ 4 \end{pmatrix}\right) \times \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} = \mathbf{0}$

b) $\left(\mathbf{r} - \begin{pmatrix} 3 \\ -7 \\ 0 \end{pmatrix}\right) \times \begin{pmatrix} 0 \\ 4 \\ -5 \end{pmatrix} = \mathbf{0}$